



# E-Commerce Ecosystem & Project Closure

Milestone Review & Production Build finalization. Delivering a wellness-centric direct ordering workspace.

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FULL-STACK INTERNSHIP PROGRAM | JUNE 2026

# Journey Recap

WEEK 1 TO WEEK 4 PROGRESS

 Shiva Refresh



## Phase 1: Week 1

Identified manual ordering bottlenecks. Discovered high customer drop-off due to slow menu browsing over unstructured chat. Defined digital catalog objectives.



## Phase 2: Weeks 2-3

Designed & built core UI in React and Tailwind CSS. Implemented product category filtering, real-time search, client-side cart management, and local storage state persistence.



## Phase 3: Week 4

Built the Chef Dashboard with passcode entry & device security visibility toggles. Finalized production bundler compliance and conducted comprehensive flow validation.

# Final System Overview



DELIVERED SCOPE VS PLANNED CORE SPECS

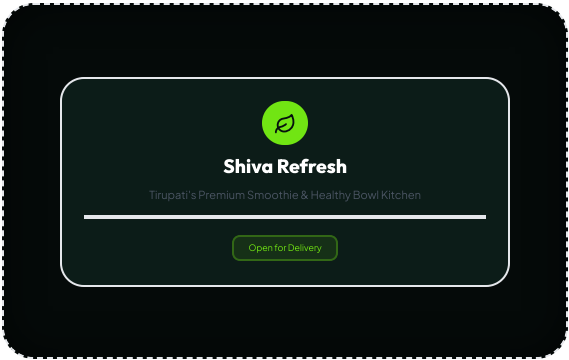
| FEATURE SPEC                | IMPLEMENTATION STRATEGY                                                        | STATUS         | IMPACT                             |
|-----------------------------|--------------------------------------------------------------------------------|----------------|------------------------------------|
| Wellness Menu Catalog       | Filtering by health category (PCOD/Weight Loss) + popup details.               | ✓100% Complete | Enhanced product discovery.        |
| Interactive Shopping Basket | Reactive cart drawer with LocalStorage caching to save current progress.       | ✓100% Complete | Zero data loss on page reload.     |
| WhatsApp API Bridge         | Compiles basket items into URI-safe text formatting and redirects on dispatch. | ✓100% Complete | Frictionless, serverless ordering. |
| Administrative Ingress      | Chef portal overview, menu updates, order histories, status switches.          | ✓100% Complete | Self-managing storefront.          |

# Customer Interface

INTUITIVE WELLNESS LANDING LAYOUT



- ✦ **Premium Branding and Aesthetics:** Dark theme color system to invoke organic, luxurious wellness feel. Uses Outfit and Manrope typography for premium polish.
- 📏 **Component Hierarchy:** Clean Hero banner, interactive category selector widgets, responsive food card grid, chef story timeline, and a quick-action footer.
- 📱 **Fully Responsive Design:** Built mobile-first to accommodate target customers order-on-the-go. Navigation drawer handles mobile menu selections smoothly.

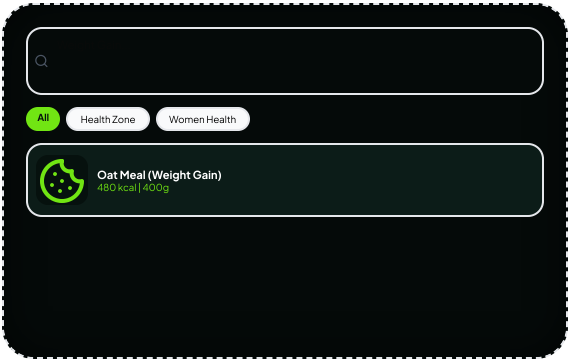


# Search & Category Filters

NAVIGATING THE HEALTH-CONSCIOUS MENU



- Real-time Fuzzy Search:** Instantly searches title, descriptions, or specific ingredients. Helps users filter out foods containing allergens.
- Diet-Specific Categories:** Quick access tabs for Health Zone, Women Health (hormonal & PCOS focus), Berry Smoothies, Dryfruit Smoothies, and Delights.
- Nutritional Quick-View Drawer:** Tapping a product details button reveals ingredients, exact serving size (ml/g), and calorie counts.



# Cart & State Persistence

ELIMINATING CHECKOUT ITERATION LOSS

- ✔ **Slide-out Cart Drawer:** Maintains visual flow. Shoppers can add multiple smoothies or health bowls while continuing to scroll and explore the menu.
- 🔄 **No-Lose Caching:** Uses custom React lifecycle tracking linked to browser 'LocalStorage'. The basket state is maintained even if browser tab crashes, reloads, or customer steps away.
- 🧮 **Automatic Calculation:** Instantly calculates item values, sets free delivery eligibility limits, and displays exact cart totals dynamically.

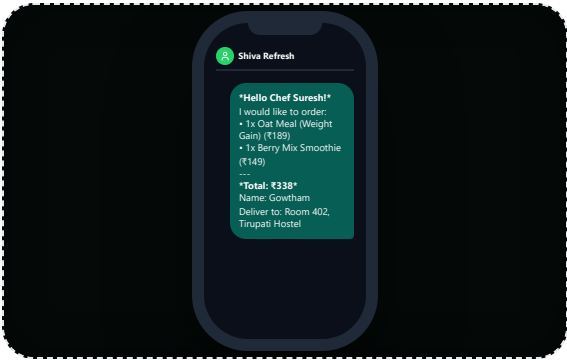


# WhatsApp Integration

THE DIRECT MERCHANT CHECKOUT ENGINE



- 🔗 **Zero-Middleman Setup:** Eliminates the cost of database servers and transaction processing by routing orders straight to the chef's WhatsApp number.
- 📄 **Prefilled WhatsApp Message Structure:** Automatically formats cart items, quantities, price calculations, and delivery details into a cleanly styled WhatsApp chat script.
- ⚡ **One-Tap Order Confirmation:** Clicking the dispatch button opens WhatsApp Web or the WhatsApp mobile app preloaded with the order content ready to send.



# Chef Portal

THE STORE ADMINISTRATION SYSTEM

- Secure Verification Guard: Hides administrative entrance behind a 5–click logo gesture or secret URL query. Requires entering the administrative passcode: 6301.
- Real-time Catalog Controls: Allows toggling item availability (In Stock / Out of Stock) instantly, making ingredients adjustments easy.
- Store Configurations: Toggles active store status between "Open" and "Closed" to pause incoming orders during rush hours.

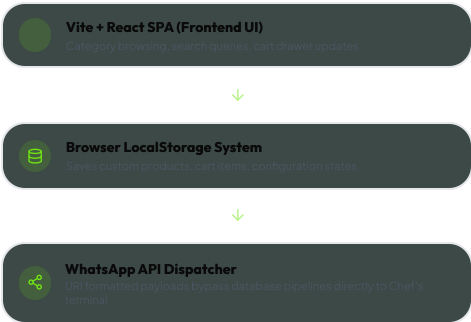


# System Architecture

CLIENT-CENTRIC SERVERLESS MODEL

Shiva Refresh

-  **Zero Backend Infrastructure:** Leverages the customer's browser engine to handle all logic. Reduces maintenance overhead to 0%.
-  **Client State Synchronization:** Hooks manage the in-memory variables. Synchronized with LocalStorage to secure data lifecycle.
-  **WhatsApp Bridge Interface:** Converts the local React order objects into standard query parameter payloads for the external WhatsApp API.



## Code Highlights

 Shiva RefreshSECURITY INGRESS STATE LOGIC IN APP.TSX

---

```
// Hidden Chef Portal visibility state handler in App.tsx
const [showChefButton, setShowChefButton] = useState() => {
  try {
    if (typeof window !== "undefined") {
      const params = new URLSearchParams(window.location.search);

      if (params.has("chef") || params.has("admin") || params.has("owner")) {
        // Returns true dynamically to show the admin lock button temporarily
        return true;
      }

      // Fallback to checking local storage if logo gesture was clicked 5 times
      return localStorage.getItem("shiva_refresh_show_chef_button") === "true";
    }
  } catch (e) {
    console.error(e);
  }
  return false;
};
```

# Database & Storage Design

SERVERLESS LOCALSTORAGE SCHEMAS



shiva\_refresh\_show\_chef\_button

BOOLEAN

Determines whether the administrative entrance lock icon remains visible on the header. Set via the 5-click logo gesture.

shiva\_refresh\_products

JSON ARRAY

Holds custom added, edited, or deleted menu products. Hydrates the active state of the store menu catalog.

shiva\_refresh\_orders

JSON ARRAY

Caches checkout orders locally with unique Order ID streams (e.g. "SR-7430"), customer addresses, times, and contents.

shiva\_refresh\_business

JSON OBJECT

Saves configuration settings: store open/closed indicator, chef description biography, and custom support dispatch numbers.

# Technical Challenges & Solutions

 Shiva Refresh

PART 1: ORDER FORMATTING & LOCAL CACHING

Challenge 1: Formatting Complex WhatsApp Message Payloads

Completed

**Problem:** Customers ordering nested items (like customized bowls and sugar-free smoothies) require structured lists with prices, subtotals, and delivery coordinates that fit WhatsApp URI constraints.  
**Solution:** Wrote a string template generator using `encodeURIComponent` with line-break character mapping (`\n`) and bold markdown formatting (`*`) to structure message summaries clearly on mobile displays.

Challenge 2: Serverless State Caching

Completed

**Problem:** Standard React states reset on page refresh. Setting up external database hosting would add monthly costs for the business owner.  
**Solution:** Built a data pipeline binding React hook lifecycle states directly to browser `LocalStorage`. Ensures the basket, menu updates, and orders remain synchronized locally for free.

# Technical Challenges & Solutions

## PART 2: INGRESS PROTECTION & URL PARAM CACHING



### Challenge 3: Shared Device Admin Visibility Exposure

Completed

**Problem:** If a chef logs in from a shared computer at the store, the Chef Portal entry button remains visible to other customers on that browser.  
**Solution:** Designed a **"Device Security"** button inside settings that wipes the persistent `'shiva_refresh_show_chef_button'` key from local storage and redirects the page, securing administrative entry instantly.

### Challenge 4: Caching of Temporary URL Query Flags

Completed

**Problem:** Accessing the app with `?admin=1` saved the permission permanently to local storage, keeping the portal visible even after removing `?admin=1` from the URL.  
**Solution:** Refactored the state initializer in `'App.tsx'` to read the search params dynamically during load. It displays the button temporarily without writing it to `localStorage`, so removing the parameter and reloading hides it.

# Testing & Verification

ENSURING APPLICATION STABILITY & CODE QUALITY



- Production Build Validation:** Executed `npm run build` checks. Verified that code compiles cleanly without TypeScript, React 19, or CSS layout errors.
- Interactive Cart Tests:** Verified add-to-order hooks, item count increments/decrements, clear cart commands, and local storage state persistence.
- Ingress Passcode Access Test:** Tested login validations. Confirmed that entering the master passcode `6301` opens dashboard tabs, while invalid passcode entry triggers error indicators.

Terminal Test Results:

```
$ tsc --noEmit (Success)
$ vite build
✓ 1732 modules transformed.
rendering chunks...
dist/index.html      1.56 kB
dist/assets/index.css 75.43 kB
dist/assets/index.js 348.27 kB
✓ built in 4.15s - Success
```

# Learnings, Limitations & Future Scope

PROJECT WRAP-UP AND EVOLUTION ROADMAP



## Key Learnings

- \* React 19 State Synchronization
- \* Client-side persistence strategies
- \* Building responsive styling frameworks in Tailwind CSS
- \* Creating dynamic ingress triggers without backends



## Limitations

- \* Caching is restricted to the browser; menu edits or order logs on one device are not visible on other devices.
- \* Offline modes require a browser cookie/local storage reset to clear space.



## Future Scope

- \* Integration of external database (Supabase or Firebase) for multi-device sync.
- \* SMS order dispatching integration.
- \* Integration of local payment gateway checkout mechanisms.



# Questions & Answers

Thank you for your time. The Shiva Refresh web ordering catalog is compiled, tested, and ready for launch.

**D. Gowtham & P. Charan Thej**

FULL-STACK DEVELOPMENT INTERNS